John S. Cost
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EDUCATION

Carnegie Mellon University Pittsburgh, PA
Biology, Computational Biology, Bachelor's and Minor May 2025

Relevant Coursework: Data Structures & Algorithms, Graph Theory, Discrete Mathematics, Computer Vision, Machine Learning, Software Engineering, Systems Programming, Git & Version Control

RESEARCH & ENGINEERING EXPERIENCE

CBG Remote, part-time
Web Developer, HTML / JavaScript Spring '24 – Present

- Build and deploy full-stack web applications for service industry operators. Client-facing UI, operator-facing custom CMS.

Benam Lung Microengineering Research Lab, University of Pittsburgh Pittsburgh, PA
Undergraduate Research Intern — ML Tooling Development June '24 - Dec 24'

- Developed ML and computer-vision systems for microscopy-based analysis of lung-on-chip experiments, saving dozens of researcher hours.
- Designed and implemented a full-stack CV pipeline to identify objects, extract quantitative metrics, and automatically generate summary figures (50× reduction in manual analysis time).
- Built a scalable motion-quantification system for 1000+ video samples using preprocessing pipelines and deep learning for pattern recognition and statistical reporting.

AI Summer Scholars Program, Carnegie Mellon University Pittsburgh, PA
Instructor / ML Mentor June '23 - Aug '23

- Taught a 5-week, instructor-led seminar on deep learning and software engineering to 25+ students.
- Mentored 6 students through full-stack ML capstone projects using TensorFlow and PyTorch.

PROJECT EXPERIENCE

Meta-Path Graph Modeling for Cell Phenotyping
Python, GNNs, Graph ML, scRNA-seq, Large Dataset
- Built heterogeneous graphs combining multiple measures of similarity (molecular/spatial).
- Designed interpretable meta-paths; flattened hetero-graphs via adj-matrix composition.
- Evaluated HAN / MetaPath2Vec for node-level phenotyping & clustering.

Microscopy Computer Vision Analysis Tool
Python, OpenCV, PyTorch, NumPy
- Solo architect/implement automated CV pipeline to detect, label, count, objects in microscopy videos.
- Extracted quantitative metrics (counts, motion, trajectories) across 1000+ samples.
- Replaced manual analysis with reproducible batch processing (~50× time reduction).
- Designed modular preprocessing, inference, and reporting stages.

TECHNICAL SKILLS

Languages: Python, C, Bash, HTML/CSS, JavaScript
Machine Learning: PyTorch, TensorFlow, NumPy, OpenCV, CNNs, RNNs, Transformers, Optical Flow, Hugging Face, Temporal Modelling
Developer Tools: Git, Agile, Technical Documentation, Jupyter, Docker
Cloud / Systems: AWS, Linux, CPU/GPU Architecture Basics